

Area & Perimeter — Full Solutions & Explanations

Digital SAT Style | Complete Working for Every Question

Q1 — EASY

■ **Correct Answer: C) 60 cm²**

STEP-BY-STEP SOLUTION:

1. Area = length $\times$ width = $12 \times 5 = 60 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $17 = 12 + 5$, adding instead of multiplying.
- ✗ B: $34 = 2(12 + 5)$, the perimeter.
- ✗ D: $120 = 2 \times 60$, doubling the area.

■ **TRAP ALERT:** Computing perimeter ($2(l+w)=34$) instead of area ($l \times w=60$).

■ **KEY INSIGHT:** Area = $l \times w$. Perimeter = $2(l+w)$. Don't confuse them.

■ **FAST METHOD:** $12 \times 5 = 60$.

Q2 — EASY

■ **Correct Answer: C) 144 cm²**

STEP-BY-STEP SOLUTION:

1. Side = $48/4 = 12 \text{ cm}$.
2. Area = $12^2 = 144 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 12 is the side length, not the area.
- ✗ B: $96 = 48 \times 2$, meaningless.
- ✗ D: $192 = 48 \times 4$, multiplying perimeter by 4.

■ **TRAP ALERT:** Reporting the side length (12) instead of the area (144).

■ **KEY INSIGHT:** Square: side = perimeter/4. Area = side².

■ **FAST METHOD:** $(48/4)^2 = 12^2 = 144$.

Q3 — EASY

■ **Correct Answer: B) 30 cm²**

STEP-BY-STEP SOLUTION:

1. Area = $(1/2) \times \text{base} \times \text{height} = (1/2)(10)(6) = 30 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $16 = 10 + 6$, adding instead of multiplying.
- ✗ C: $60 = 10 \times 6$, forgetting the $(1/2)$.
- ✗ D: $120 = 2 \times 60$, even worse.

■ **TRAP ALERT:** Forgetting the $(1/2)$ in triangle area formula. Area = $(1/2)bh$, not bh .

■ **KEY INSIGHT:** Triangle area = $(1/2)(\text{base})(\text{height})$. Always halve the rectangle.

■ **FAST METHOD:** $(1/2)(10)(6) = 30$.

Q4 — EASY

■ **Correct Answer: C) 154 cm²**

STEP-BY-STEP SOLUTION:

1. $\text{Area} = \pi r^2 = (22/7)(7^2) = (22/7)(49) = 22 \times 7 = 154 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $22 = \pi d/2$, confusion with circumference.

✗ B: $44 = 2\pi r$, the circumference.

✗ D: $308 = 2 \times 154$, doubling the area.

■ **TRAP ALERT:** Using $2\pi r$ (circumference) instead of πr^2 (area), or using diameter instead of radius.

■ **KEY INSIGHT:** Area = πr^2 . Circumference = $2\pi r$. Don't mix them up.

■ **FAST METHOD:** $(22/7)(49) = 154$.

Q5 — EASY

■ **Correct Answer: D) 8**

STEP-BY-STEP SOLUTION:

1. $\text{Perimeter} = 2(l+w) = 2((x+3)+4) = 2(x+7) = 30$.

2. $x+7 = 15 \rightarrow x = 8$.

3. Check: length=11, width=4. $\text{Perimeter} = 2(11+4) = 30$ ✓

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $x=4 \rightarrow P=2(11)=22$.

✗ B: $x=5 \rightarrow P=2(12)=24$.

✗ C: $x=7 \rightarrow P=2(14)=28$.

■ **TRAP ALERT:** Setting up $l \times w = 30$ (area formula) instead of $2(l+w) = 30$ (perimeter).

■ **KEY INSIGHT:** Perimeter = $2(l+w)$. Set up the equation carefully.

■ **FAST METHOD:** $2(x+7) = 30 \rightarrow x = 8$.

Q6 — EASY

■ **Correct Answer: D) 16 cm²**

STEP-BY-STEP SOLUTION:

1. Area of larger = $5^2 = 25 \text{ cm}^2$.

2. Area of smaller = $3^2 = 9 \text{ cm}^2$.

3. Difference = $25 - 9 = 16 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $2 = 5 - 3$, subtracting sides instead of areas.

✗ B: $4 = 2^2$, squaring the side difference.

✗ C: $8 = 2 \times 4$, arbitrary.

■ **TRAP ALERT:** Subtracting side lengths ($5 - 3 = 2$) instead of areas ($25 - 9 = 16$).

■ **KEY INSIGHT:** Area difference $\neq$ side difference. Always compute areas separately, then subtract.

■ **FAST METHOD:** $5^2 - 3^2 = 25 - 9 = 16$.

Q7 — MEDIUM

■ **Correct Answer: B) 50 cm²**

STEP-BY-STEP SOLUTION:

1. $\text{Area} = (1/2)(b_1 + b_2)(h) = (1/2)(8 + 12)(5) = (1/2)(20)(5) = 50 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $40 = 8 \times 5$, using only one base.

✗ C: $60 = 12 \times 5$, using only the longer base.

✗ D: $100 = 20 \times 5$, forgetting the $(1/2)$.

■ **TRAP ALERT:** Forgetting the $(1/2)$, or using only one base instead of the average of both bases.

■ **KEY INSIGHT:** Trapezoid area = $(1/2)(\text{sum of parallel sides})(\text{height})$.

■ **FAST METHOD:** $(1/2)(20)(5) = 50$.

Q8 — MEDIUM

■ **Correct Answer: B) 62.8 cm**

STEP-BY-STEP SOLUTION:

1. $C = \pi d = 3.14 \times 20 = 62.8 \text{ cm}$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $31.4 = \pi r$, using radius formula for circumference without the 2.

✗ C: $125.6 = \pi d \times 2$, doubling unnecessarily.

✗ D: $314 = \pi r^2$, this is the area (in wrong units too).

■ **TRAP ALERT:** Using radius (10) in $C = \pi d$ formula, getting $C = 31.4$. Remember: $C = \pi d$ or $C = 2\pi r$.

■ **KEY INSIGHT:** $C = \pi d = 2\pi r$. Use diameter directly when given.

■ **FAST METHOD:** $\pi \times 20 = 62.8$.

Q9 — MEDIUM

■ **Correct Answer: B) 4**

STEP-BY-STEP SOLUTION:

1. Original area = πr^2 . New area = $\pi(2r)^2 = 4\pi r^2$.

2. Factor = $4\pi r^2 / \pi r^2 = 4$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 2 applies to circumference (linear), not area (quadratic).

✗ C: 6 has no mathematical basis here.

✗ D: 8 applies to volume when radius is doubled in 3D.

■ **TRAP ALERT:** Thinking area doubles when radius doubles. Area depends on r^2 , so it quadruples.

■ **KEY INSIGHT:** Area scales with r^2 . Double $r \rightarrow$ area $\times 4$. Triple $r \rightarrow$ area $\times 9$.

■ **FAST METHOD:** Scale factor for area = $2^2 = 4$.

Q10 — MEDIUM

■ **Correct Answer: C) 116 m²**

STEP-BY-STEP SOLUTION:

1. Path is 2m wide on EACH side, so add 4 to each dimension.

2. Outer: $(15+4) \times (10+4) = 19 \times 14 = 266 \text{ m}^2$.

3. Inner: $15 \times 10 = 150 \text{ m}^2$.

4. Path area = $266 - 150 = 116 \text{ m}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 56 uses wrong outer dimensions.

✗ B: 92 adds only 2 to each dimension instead of 4.

✗ D: 124 from computation error.

■ **TRAP ALERT:** Adding only 2 (not 4) to each dimension. Path is 2m wide on BOTH sides.

■ **KEY INSIGHT:** With a path of width w around a rectangle: $\text{outer} = (l+2w)(w+2w)$. Path area = outer – inner.

■ **FAST METHOD:** $(15+4)(10+4) - 150 = 266 - 150 = 116 \text{ m}^2$.

Q11 — MEDIUM

■ **Correct Answer: B) 77 cm^2**

STEP-BY-STEP SOLUTION:

1. Radius = $14/2 = 7 \text{ cm}$.

2. Full circle area = $\pi r^2 = (22/7)(49) = 154 \text{ cm}^2$.

3. Semicircle area = $154/2 = 77 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $38.5 = 77/2$, quartering the circle instead of halving.

✗ C: 154 is the full circle area.

✗ D: $308 = 2 \times 154$, doubling instead of halving.

■ **TRAP ALERT:** Using the diameter as radius: $\pi(14)^2$ gives $4 \times$ the correct area.

■ **KEY INSIGHT:** Semicircle = half the full circle. Make sure to use RADIUS, not diameter.

■ **FAST METHOD:** $(1/2)(22/7)(7^2) = 77$.

Q12 — MEDIUM

■ **Correct Answer: A) $25\pi \text{ cm}^2$**

STEP-BY-STEP SOLUTION:

1. Sector area = $(\theta/360^\circ) \times \pi r^2 = (90/360) \times \pi(100) = (1/4)(100\pi) = 25\pi \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ B: $50\pi = (1/2)\pi r^2$, using semicircle instead of quarter.

✗ C: $75\pi = (3/4)\pi r^2$, using the wrong fraction.

✗ D: 100π is the full circle area.

■ **TRAP ALERT:** Using $(\theta/180^\circ)$ instead of $(\theta/360^\circ)$.

■ **KEY INSIGHT:** Sector area = $(\text{angle}/360^\circ) \times \pi r^2$. Think of it as a fraction of the full circle.

■ **FAST METHOD:** $90^\circ/360^\circ = 1/4$. Area = $(1/4)(100\pi) = 25\pi$.

Q13 — MEDIUM

■ **Correct Answer: B) $16\pi \text{ cm}^2$**

STEP-BY-STEP SOLUTION:

1. Area = $\pi(R^2 - r^2) = \pi(25 - 9) = 16\pi \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $8\pi = \pi(R-r)^2 \times 2$, wrong formula.

✗ C: $25\pi = \pi R^2$, the full outer circle.

✗ D: $34\pi = \pi(R^2+r^2)$, adding instead of subtracting.

■ **TRAP ALERT:** Computing $\pi(R-r)^2 = \pi(4) = 4\pi$ instead of $\pi(R^2-r^2) = 16\pi$.

■ **KEY INSIGHT:** Ring area $= \pi(R^2-r^2)$. Subtract the areas, don't subtract the radii first.

■ **FAST METHOD:** $\pi(25-9) = 16\pi$.

Q14 — HARD

■ **Correct Answer: C) 44%**

STEP-BY-STEP SOLUTION:

1. New side $= 1.2s$. New area $= (1.2s)^2 = 1.44s^2$.

2. Increase $= 1.44 - 1 = 0.44 = 44\%$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 20% treats area as scaling linearly with side.

✗ B: $40\% = 2 \times 20\%$, adding percent changes.

✗ D: 120% is the new area as percent of original, not the increase.

■ **TRAP ALERT:** Thinking area increases by 20% when sides increase by 20%. Percent changes **MULTIPLY**, not add.

■ **KEY INSIGHT:** When a dimension changes by factor k , area changes by k^2 . $1.2^2 = 1.44 \rightarrow 44\%$ increase.

■ **FAST METHOD:** $(1.2)^2 - 1 = 0.44 = 44\%$.

Q15 — HARD

■ **Correct Answer: D) 1800 m²**

STEP-BY-STEP SOLUTION:

1. Let width $= w$, length $= L$. Fencing: $2w + L = 120$.

2. $L = 120 - 2w$. Area $= wL = w(120 - 2w) = 120w - 2w^2$.

3. Maximum at $w = -120/(2 \times (-2)) = 30$.

4. $L = 120 - 60 = 60$. Area $= 30 \times 60 = 1800 \text{ m}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $900 = 30^2$, treating it as a square.

✗ B: 1200 uses $w=20$, $L=80$.

✗ C: 1500 uses $w=25$, $L=70$.

■ **TRAP ALERT:** Using $2w+2L=120$ (4 sides) instead of $2w+L=120$ (3 sides, barn provides one side).

■ **KEY INSIGHT:** Optimization: for 3-sided fencing with total P , max area at $w=P/4$, $L=P/2$.

■ **FAST METHOD:** $w=30$, $L=60$. Area $= 1800$. ■ **Desmos:** graph $A=w(120-2w)$, find maximum.

Q16 — HARD

■ **Correct Answer: B) 154 cm²**

STEP-BY-STEP SOLUTION:

1. Side of square $= \sqrt{196} = 14 \text{ cm}$.

2. Inscribed circle has diameter = side = 14, so radius = 7.

3. Circle area = $\pi(7^2) = (22/7)(49) = 154 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 77 is the semicircle area.

✗ C: 196 is the square area, not the circle.

✗ D: $616 = \pi(14^2)$, using side as radius.

■ **TRAP ALERT:** Using the side (14) as the radius instead of the diameter. Radius = side/2.

■ **KEY INSIGHT:** Circle inscribed in square: diameter = side of square. Radius = side/2.

■ **FAST METHOD:** Side=14 $\rightarrow$ r=7 $\rightarrow$ area = $(22/7)(49) = 154$.

Q17 — HARD

■ **Correct Answer:** C) 36 cm

STEP-BY-STEP SOLUTION:

1. Hypotenuse = $\sqrt{81+144} = \sqrt{225} = 15 \text{ cm}$.

2. Perimeter = $9 + 12 + 15 = 36 \text{ cm}$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $21 = 9+12$, forgetting the hypotenuse.

✗ B: $30 = 2 \times 15$, doubling the hypotenuse.

✗ D: $45 = 3 \times 15$, tripling it.

■ **TRAP ALERT:** Forgetting to find the hypotenuse before computing the perimeter.

■ **KEY INSIGHT:** 9-12-15 = $3 \times (3-4-5)$ Pythagorean triple. Always find all three sides before adding.

■ **FAST METHOD:** Recognize $3 \times (3-4-5) \rightarrow$ hypotenuse=15. $P=9+12+15=36$.

Q18 — HARD

■ **Correct Answer:** B) 24

STEP-BY-STEP SOLUTION:

1. Base on x-axis from (0,0) to (8,0): base = 8.

2. Height = y-coordinate of (4,6) = 6.

3. Area = $(1/2)(8)(6) = 24$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 16 uses wrong height or base.

✗ C: 32 uses $(1/2)(8)(8)$.

✗ D: $48 = 8 \times 6$, forgetting the $(1/2)$.

■ **TRAP ALERT:** Forgetting $(1/2)$ in the area formula.

■ **KEY INSIGHT:** If base is on an axis, height = perpendicular coordinate of the opposite vertex.

■ **FAST METHOD:** $(1/2)(8)(6) = 24$. ■ **Desmos:** plot points, measure.

Q19 — HARD

■ **Correct Answer:** B) 75 cm^2

STEP-BY-STEP SOLUTION:

1. Let width = w. Length = 3w.

2. $2(w+3w) = 40 \rightarrow 8w = 40 \rightarrow w = 5$.
3. Length = 15. Area = $5 \times 15 = 75 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 48 uses $w=4$.
- ✗ C: $100 = (\text{perimeter}/4)^2$.
- ✗ D: 120 uses wrong dimensions.

■ **TRAP ALERT:** Setting up $2(w+w)=40$ instead of $2(w+3w)=40$.

■ **KEY INSIGHT:** Translate "3 times its width" to $l=3w$, then use perimeter formula.

■ **FAST METHOD:** $8w=40 \rightarrow w=5$. Area $= 5 \times 15 = 75$.

Q20 — HARD

■ **Correct Answer:** B) $16\sqrt{3} \text{ cm}^2$

STEP-BY-STEP SOLUTION:

1. Equilateral triangle area = $(\sqrt{3}/4)s^2$.
2. $= (\sqrt{3}/4)(64) = 16\sqrt{3} \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $12\sqrt{3}$ uses $s=4\sqrt{3}$, wrong input.
- ✗ C: $24\sqrt{3}$ uses $(\sqrt{3}/4)(96)$, wrong s^2 .
- ✗ D: $32\sqrt{3} = (\sqrt{3}/2)s^2$, wrong formula.

■ **TRAP ALERT:** Using $(\sqrt{3}/2)s$ instead of $(\sqrt{3}/4)s^2$ — mixing up height and area formulas.

■ **KEY INSIGHT:** Equilateral triangle: area = $(\sqrt{3}/4)s^2$. Height = $(\sqrt{3}/2)s$.

■ **FAST METHOD:** $(\sqrt{3}/4)(64) = 16\sqrt{3}$.

Q21 — HARD

■ **Correct Answer:** B) $44\pi \text{ m}^2$

STEP-BY-STEP SOLUTION:

1. Outer radius = $10+2 = 12 \text{ m}$.
2. Path area = $\pi(12^2) - \pi(10^2) = \pi(144-100) = 44\pi \text{ m}^2$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $40\pi = \pi(2)(20)$, wrong formula.
- ✗ C: 56π from wrong outer radius.
- ✗ D: 100π is the inner circle area, not the path.

■ **TRAP ALERT:** Computing $\pi(12-10)^2 = 4\pi$ instead of $\pi(12^2-10^2) = 44\pi$.

■ **KEY INSIGHT:** Ring area = $\pi(R^2-r^2)$. Subtract areas, not radii.

■ **FAST METHOD:** $\pi(144-100) = 44\pi$.

Q22 — HARD

■ **Correct Answer:** B) 6×8

STEP-BY-STEP SOLUTION:

1. $l \times w = 48$ and $2(l+w) = 28 \rightarrow l+w = 14$.
2. Solve: w and l are roots of $t^2-14t+48=0$.

3. $(t-6)(t-8)=0 \rightarrow t=6,8$. Dimensions: 6×8 .

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 4×12 : perimeter = $2(16) = 32 \neq 28$.

✗ C: 3×16 : perimeter = $2(19) = 38 \neq 28$.

✗ D: 2×24 : perimeter = $2(26) = 52 \neq 28$.

■ **TRAP ALERT:** Finding dimensions that satisfy area but not checking the perimeter.

■ **KEY INSIGHT:** When both area and perimeter are given, check BOTH constraints.

■ **FAST METHOD:** Test each option: $6+8=14 \rightarrow$ perimeter=28 ✓ and $6 \times 8=48$ ✓.

Q23 — HARD

■ **Correct Answer: C) $54\sqrt{3} \text{ cm}^2$**

STEP-BY-STEP SOLUTION:

1. Area = $(3\sqrt{3}/2)s^2 = (3\sqrt{3}/2)(36) = 54\sqrt{3} \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $27\sqrt{3}$ uses $(3\sqrt{3}/4)s^2$ (wrong formula).

✗ B: $36\sqrt{3} = (\sqrt{3})s^2$, wrong coefficient.

✗ D: $72\sqrt{3} = (2\sqrt{3})s^2$, wrong formula.

■ **TRAP ALERT:** Using the wrong coefficient. Regular hexagon = 6 equilateral triangles.

■ **KEY INSIGHT:** Regular hexagon area = $(3\sqrt{3}/2)s^2$. Alternatively = $6 \times$ equilateral triangle area.

■ **FAST METHOD:** $(3\sqrt{3}/2)(36) = 54\sqrt{3}$.

Q24 — HARD

■ **Correct Answer: A) Circle, by approximately 33 cm^2**

STEP-BY-STEP SOLUTION:

1. Square: side = $44/4 = 11$. Area = 121 cm^2 .

2. Circle: $C = 2\pi r = 44 \rightarrow r = 44/(2 \times 22/7) = 44 \times 7/44 = 7$. Area = $(22/7)(49) = 154 \text{ cm}^2$.

3. Difference = $154 - 121 = 33 \text{ cm}^2$. Circle is larger.

WHY EACH WRONG ANSWER IS WRONG:

✗ B: Square has LESS area for the same perimeter.

✗ C: 56 cm^2 from wrong calculation.

✗ D: Square is never larger than circle for same perimeter.

■ **TRAP ALERT:** Assuming the square has more area. For the same perimeter, a circle always has the largest area.

■ **KEY INSIGHT:** Isoperimetric inequality: among all shapes with the same perimeter, the circle has the largest area.

■ **FAST METHOD:** Square area=121, circle area=154, difference=33 cm^2 .

Q25 — HARD

■ **Correct Answer: A) 63 cm^2**

STEP-BY-STEP SOLUTION:

1. Parallelogram area = base $\times$ height = $14 \times 9 = 126 \text{ cm}^2$.

2. Triangle area = $(1/2)(14)(9) = 63 \text{ cm}^2$.

3. Remaining = $126 - 63 = 63 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ B: 72 uses wrong dimensions.

✗ C: $81 = 9^2$, wrong formula.

✗ D: 126 is the full parallelogram, before cutting.

■ **TRAP ALERT:** Forgetting that a triangle with the same base and height is exactly **HALF** the parallelogram.

■ **KEY INSIGHT:** Parallelogram area = bh . Triangle area = $(1/2)bh$. Triangle = half the parallelogram.

■ **FAST METHOD:** Remaining = $bh - (1/2)bh = (1/2)bh = 63$.

TOPIC SUMMARY — Area & Perimeter

Common Mistakes (Top 3)

- Forgetting $(1/2)$ in triangle and trapezoid area formulas.
- Using diameter instead of radius in circle formulas (area off by $4\times$).
- Percent changes MULTIPLY, not add: 20% increase in each side $\rightarrow$ 44% increase in area, not 40%.

Must Remember (Top 3 Rules)

- Rectangle: $A=lw$, $P=2(l+w)$. Triangle: $A=(1/2)bh$. Circle: $A=\pi r^2$, $C=2\pi r$.
- Trapezoid: $A=(1/2)(b_1+b_2)h$. Ring: $A=\pi(R^2-r^2)$.
- Equilateral triangle: $A=(\sqrt{3}/4)s^2$, $h=(\sqrt{3}/2)s$.

Expected on DSAT

- 2–3 questions per test. Common: composite figures, rings, percent changes in dimensions.
- ■ Desmos: graph area optimization problems to find the maximum quickly.

End of Solutions — Area & Perimeter